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Assignment → 3

SECTION → A

1) What is SQL? Why is it important?

→ SQL (Structured Query Language) is used to store, retrieve, and manage data in relational databases. It is important because it allow us to interact with databases efficiently, perform operations like inserting, updating, deleting and retrieving data.

2) Difference b/w DDL & DML?

→ DDL (Data Definition Language): Used to define structure (CREATE, ALTER, DROP)

DML (Data Manipulation Language): Used to manipulate data (INSERT, UPDATE, DELETE)

In simple terms:
DDL → Structure, DML → data

3) What is a Join? Types:

→ JOIN is used to combine data from multiple tables based on a common column.

Types:

* INNER JOIN → only matching records:
* LEFT JOIN → all left + matching right.
* RIGHT JOIN → all right + matching left.
* FULL JOIN → all records from both tables.

4. GROUP BY and HAVING:
* GROUP BY :- Groups rows with same values.
* HAVING :- Filters grouped data.

Eg → count students per department and filter those > 10.

5. What is Subquery?
→ A subquery is a query inside another query used to perform complex operations.

6. What is Relational Algebra?
→ It is a procedural query language used in DBMS with operations like:
SELECT, PROJECT, JOIN.

7. WHERE vs HAVING?
* WHERE → Filters before grouping.
* HAVING → filters after grouping.

SELECT SECTION - B

1) Students with marks > 75.

→
```
SELECT * FROM Students
WHERE marks > 75;
```

2) JOIN query (Student + Course)

→
```
SELECT s.name, c.course_name
FROM students s
INNER JOIN Courses c
ON s.course_id = c.course_id;
```

3| Count students per department

→ SELECT department, COUNT(*) AS total_students
FROM students
GROUP BY department;

SECTION: C

1. Employees in specific department

→ SELECT e.name
FROM Employees e
JOIN Departments d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'HR';

2. Total sales per customer

→ SELECT customer_id, SUM(amount) AS total_sales
FROM orders
GROUP BY customer_id;

3. Top 3 students

SELECT * FROM students
ORDER BY marks DESC
LIMIT 3;

SECTION → D

Example Query (Real-life: online store)

→ SELECT product_id, SUM (quantity) AS total_sold
FROM Sales
GROUP BY product_id;

* This helps businesses identify best-selling products.

JOIN Use case:

SELECT o.order_id, c.customer_name
FROM Orders o
JOIN customers c
ON o.customer_id = c.customer_id;

* Used to link related data across tables.

GROUP BY Explanation

SELECT department, AVG (salary)
FROM Employees
GROUP BY department;

* Output shows average salary per department.

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SECTION → A

1. What is DBMS? Why required?

→ DBMS is software used to store, manage and retrieve data efficiently.
It is required for security, consistency, and reduced redundancy.

2. DBMS vs File system.

→ DBMS is better because:—
* Reduces redundancy.
* Avoids inconsistency.
* Provides security.
* Supports backup & recovery.

3. Data Abstraction.

→ ~~Hig~~ Hiding internal details of data.

Levels:—
* Physical:— how data stored.
* Logical:— structure.
* View:— user view.

4. Three-Schema Architecture.

* Internal level
* Conceptual level
* External level

Important because it provides data independence

5. Data Independence.

Ability to change schema without affecting other levels.

* Makes system flexible.

6| ER Model

Graphical representation of :-

* Entities
* Attributes
* Relationships

7| Integrity constraints

Rules to maintain data accuracy:-

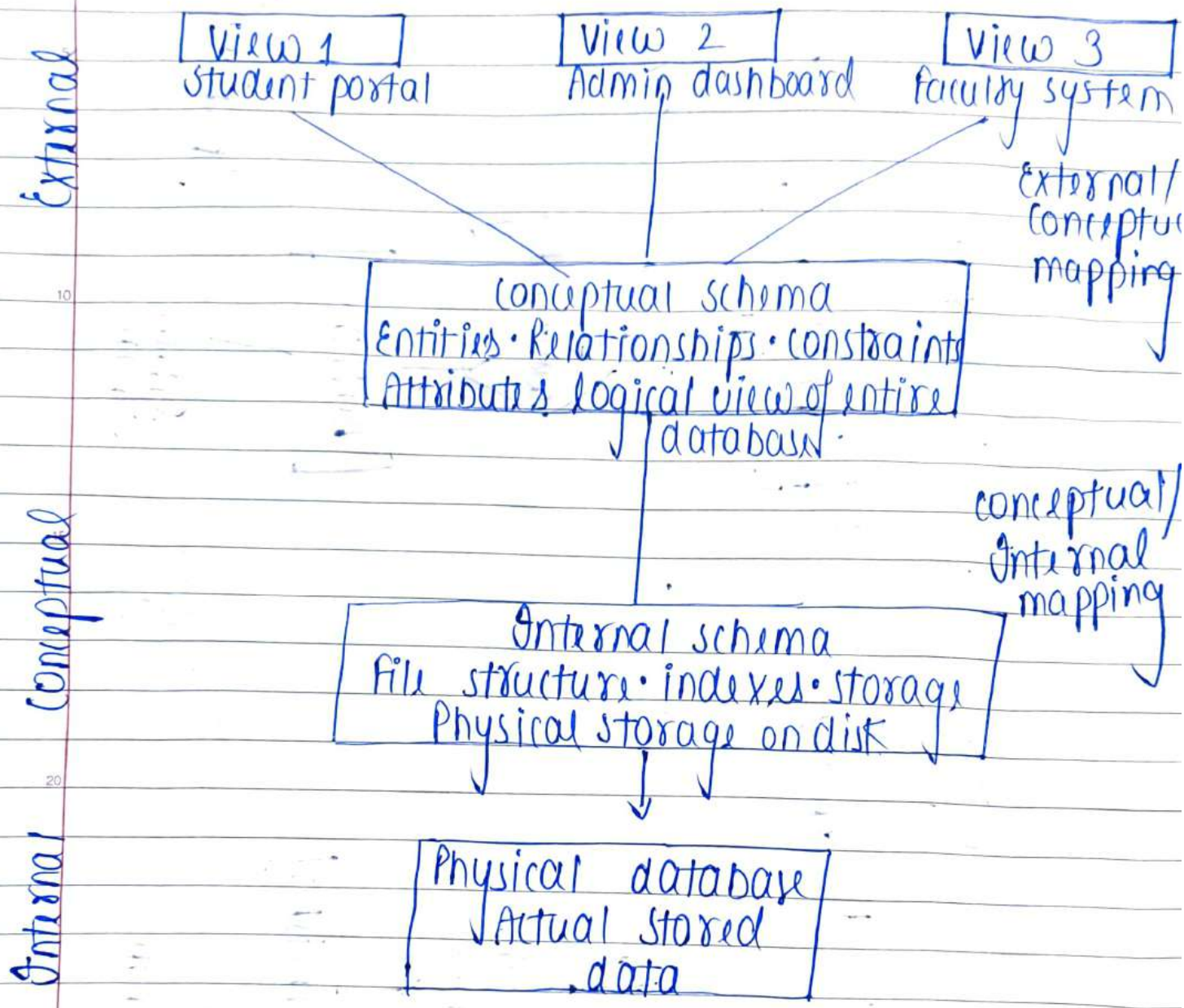
* Primary Key.
* Foreign Key.
* Unique.
* NOT NULL.

SECTION → B

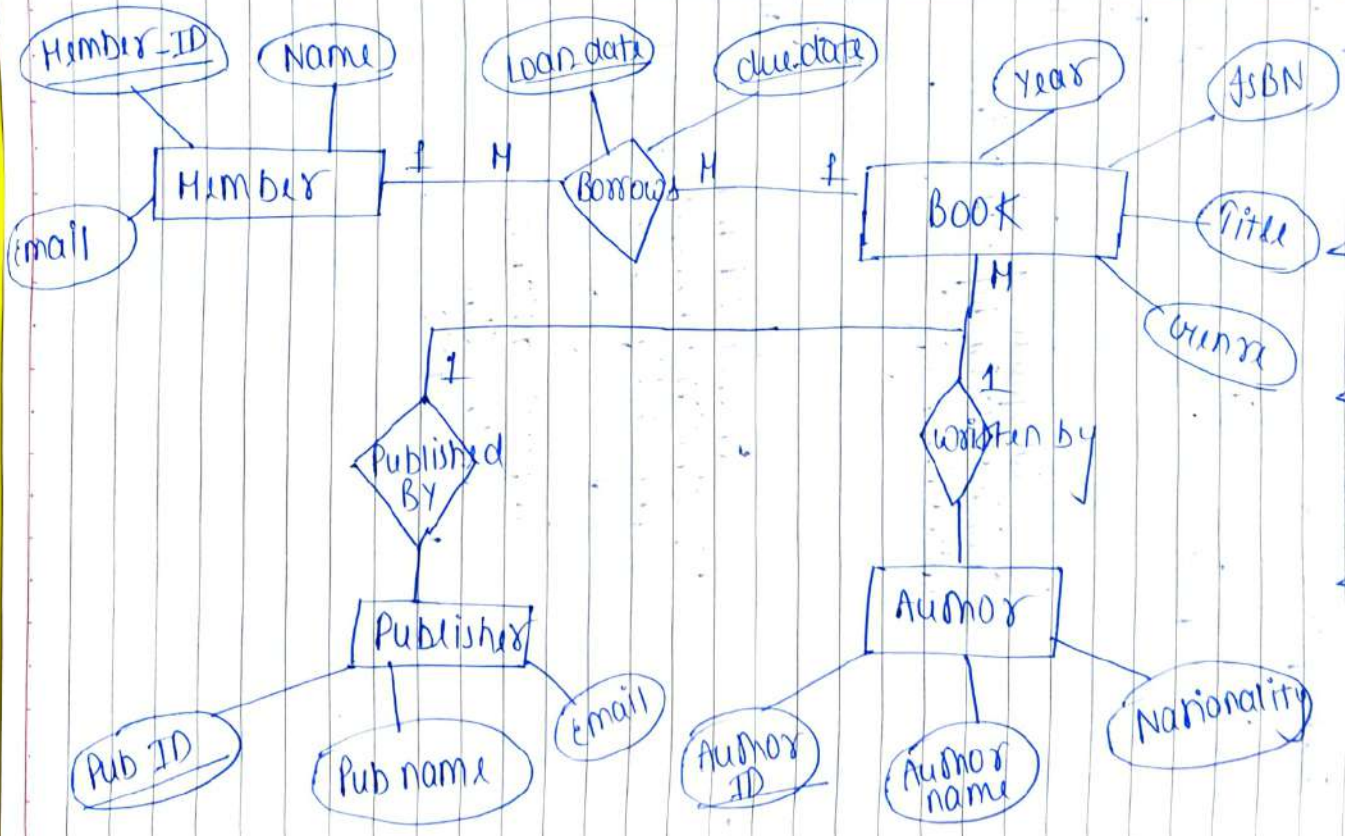
THREE-Schema Architecture: → It separates a database system into three abstraction levels, ensuring data independence.

* The external level is what individual users or application see - each user group can have a customized view.
* The conceptual level is the unified logical description of the entire database - entities, relationships and constraints, independent of physical storage.

* The internal level describes how data is actually stored - file organization, indexes, and physical access paths.



Library Management System :-



Student - Course Registration

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SECTION - C

Hospital system

Entities :-

i/ Patient ii/ Doctor iii/ Appointment

Relationships :-

* Patient books Appointment.
* Doctor attends Appointment.

E-commerce Keys

* Customer → customer_id (Primary Key (PK)).
* Product → product_id (PK)
* Order → order_id (PK).

FK :-

* order. customer_id.
* order. product_id.

Data Abstraction Example

* Admin → Full access
* Staff → limited access
* User → View only.

Section → D

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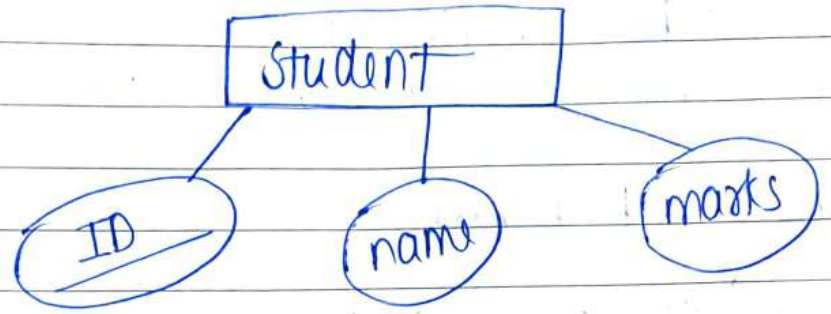
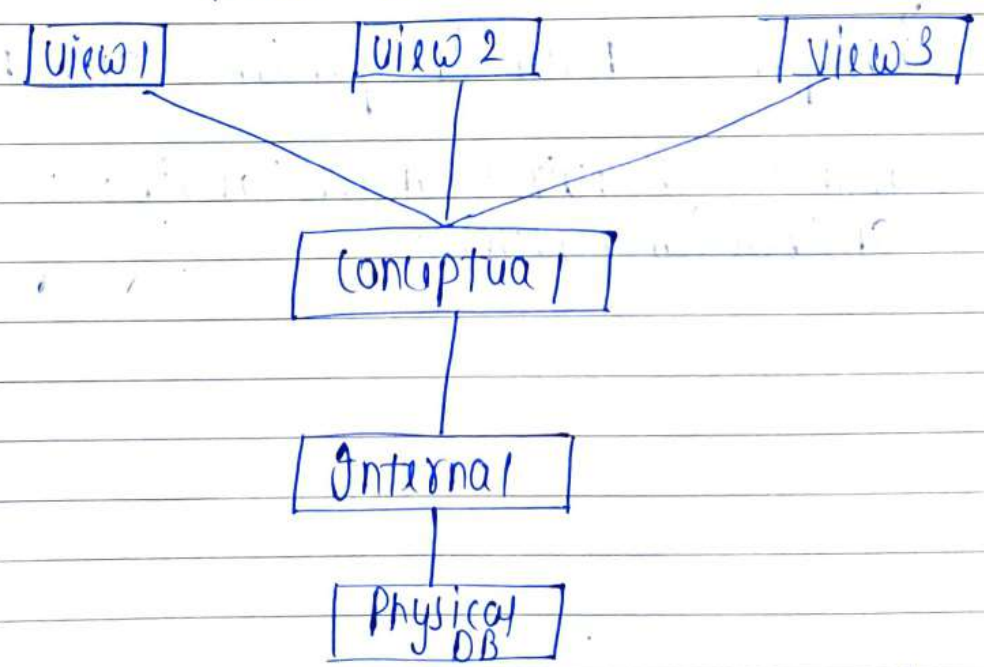


Design decision :— id is unique to identify students.

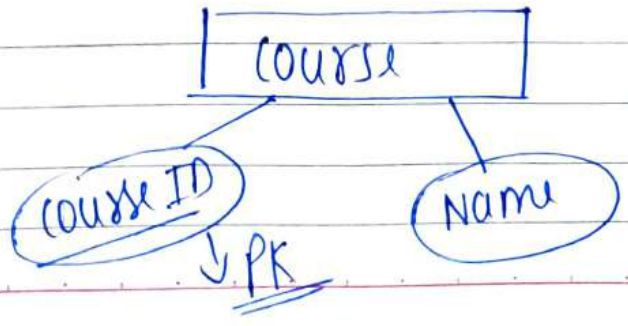
#

Three Schema



Benefit of Three :— Allows changes without affecting users.

#



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SECTION → E

Assumptions:—

1/ Each student has unique ID:
2/ One course have many students.

limitation of ER model.

* does not handle complex logic easily.

Difficult topic:—

Example :— Data Independence and more topics.

reason :— Requires understanding of multiple abstraction levels.